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Explainable Multimodal Skin Lesion Classification via Supervised Contrastive Learning and Attention-Based Fusion

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Abstract

- ✔ Skin-lesion classification is challenging because lesion categories can look similar and clinical datasets are highly imbalanced.
- ✔ FusionClassifier combines a Swin-Small Transformer, supervised contrastive learning, clinical metadata, and attention-based fusionGrad-CAM and Grad-CAM++ provide visual explanations of image-based predictions.
- ✔ On HAM10015, the paper reports 98.17% accuracy and 97.28% F1-score, with high ROC-AUCFive-fold evaluation and ablation experiments assess the contribution of the framework's components.

Problem

Visual similarity + severe class imbalance

Approach

Contrastive image learning + metadata attention

Interpretability

Grad-CAM and Grad-CAM++

Reported metrics are from the submitted paper.

Introduction

- Dermatoscopic images support the assessment of skin lesions, but visual overlap can make diagnosis difficult.
- HAM10015/HAM10000 is used in the paper as a benchmark with 10,015 images across seven diagnostic categories.
- The dataset is strongly imbalanced: melanocytic nevi dominate, while dermatofibroma and vascular lesions are rare.
- Image-only predictions may miss useful context available in age, sex, and anatomical site.

Challenge 1

Visually similar lesion classes

Challenge 2

Unequal class frequencies

Challenge 3

Need for interpretable predictions

Motivation & Objectives

Motivation

Standard classification may not explicitly separate visually similar classesSimple fusion can also treat image and metadata features as equally informative.

Overall objective

Develop an explainable multimodal framework that learns discriminative image representations and adaptively combines them with clinical metadata.

Specific objectives

1. Use supervised contrastive learning with Swin-Small to improve visual representation learning.
2. Fuse image and clinical metadata using learned attention weights.
3. Evaluate with stratified group five-fold cross-validation and component ablations.
4. Use Grad-CAM and Grad-CAM++ to inspect image regions influencing predictions.

RESEARCH QUESTION

TBA

Literature Review

Transformer models

ViT and Swin Transformer offer strong global and hierarchical image representations.

Contrastive learning

Pulls same-class examples together and separates different classes in embedding space.

Multimodal learning

Clinical metadata can complement image features; attention can adapt modality importance.

Research gaps identified by the paper

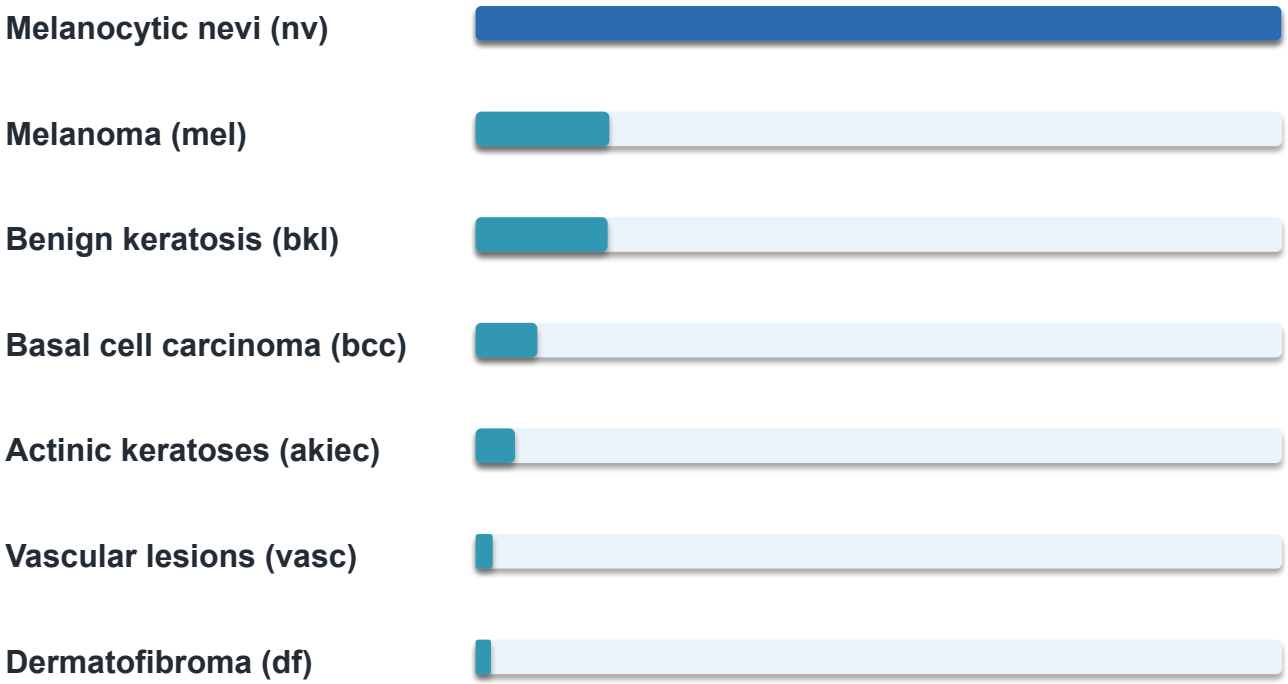
1) Discriminative representation learning is not always explicitly optimized for similar lesion categories.
2) Some fusion strategies rely on direct concatenation rather than case-specific modality weighting.
3) Explainability methods and multimodal fusion are often studied separately.
4) The proposed work combines Swin-Small + SupCon + attention fusion + dual visual explanation.

Representative references: Esteva et al(2017); Liu et al(2021); Khosla et al(2020); Ou et al(2022).

Dataset and Class Distribution

HAM10015 / HAM10000 benchmark

10,015 dermatoscopic images • 7 classes



Metadata

Age • Sex

Anatomical site

Imbalance

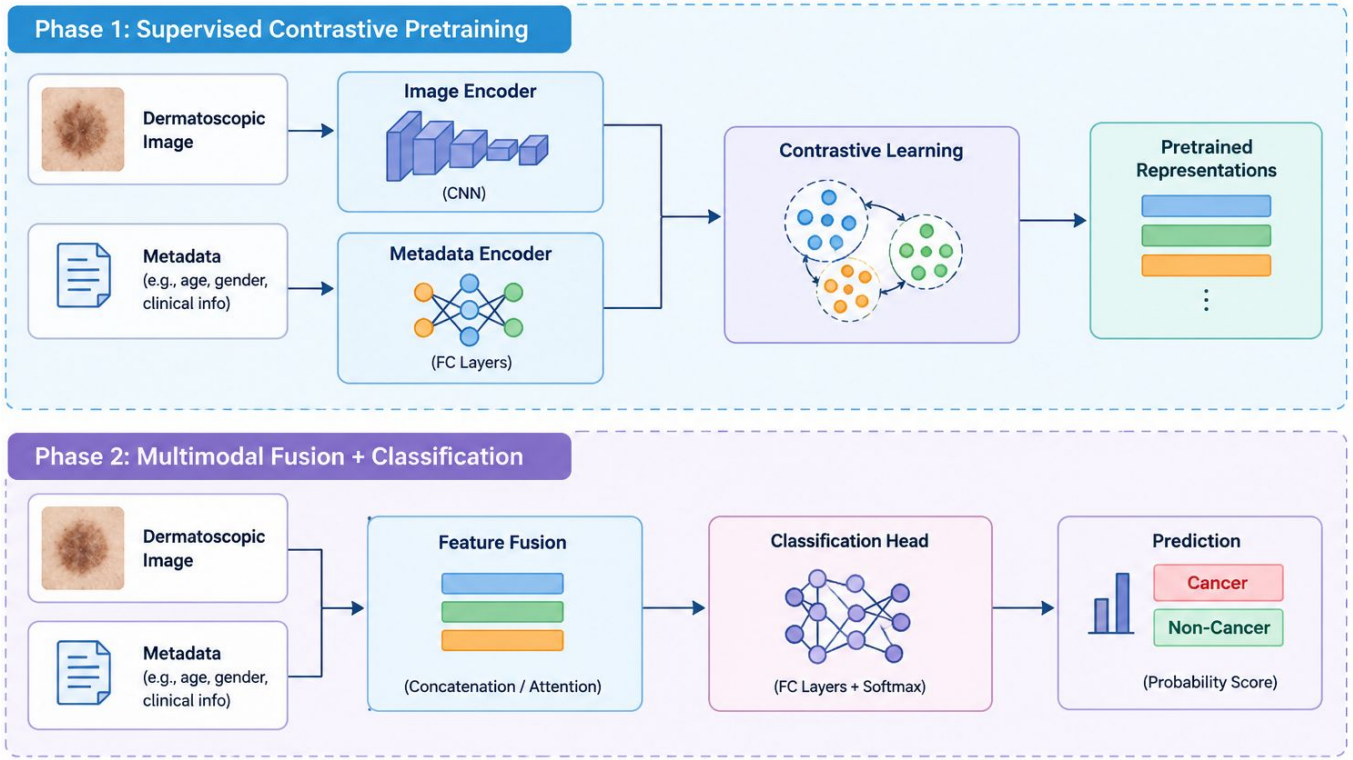
Nevi: 66.95%

DF: 1.15%

Preprocessing: resize to 224×224, ImageNet normalization, augmentation, and two views per image for SupCon.

Proposed Methodology: Two-Phase Framework

FusionClassifier learns image representations first, then integrates clinical metadata.



Evaluation: stratified group 5-fold cross-validation; lesion-associated samples remain within one fold.

Figure 01: The proposed two-phase training strategy. Phase 1 focuses on supervised contrastive pretraining, and Phase 2 combines multimodal fusion with classification.

Phase 1: Supervised Contrastive Learning

Augmented views
Two views per training image

Swin-Small encoder
768-dimensional features

Projection head
128-dimensional embeddings

Training objective

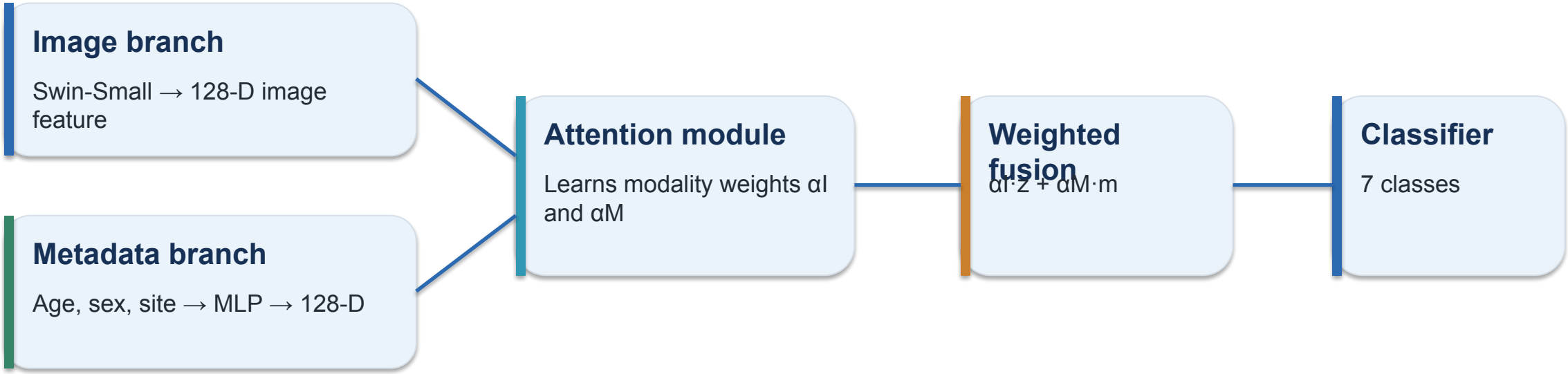
I. Same-diagnosis samples form positive pairs.
II. Different-diagnosis samples act as negatives.
III. Temperature parameter: $\tau = 0.07$.

Why SupCon?

Encourages intra-class compactness and inter-class separation, helping distinguish visually similar lesion categories.

Reported ablation: Swin-Small baseline 73.58% → 98.21% accuracy after SupCon pretraining

Phase 2: Attention-Based Multimodal Fusion



Unlike simple concatenation, attention learns how much each modality should contribute for a given sample.

Loss: class-weighted cross-entropy to mitigate the severe class imbalance.

Experimental Setup and Evaluation

Implementation

PyTorch • Kaggle • NVIDIA Tesla P100 GPU

Training schedule

SupCon: 80 epochs

Multimodal: 50 epochs/fold

Optimization

Adam • learning rate 10^{-4}

Class-weighted cross-entropy

Validation and metrics

a. Stratified Group K-Fold cross-validation (5 folds).

b. Lesion-associated samples kept in a single fold.

c. Augmentation: crops, flips, rotation, blur, color/brightness changes.

a) Accuracy, precision, recall, F1-score and specificity.

b) Macro-averaged ROC-AUC.

c) Grad-CAM and Grad-CAM++ for qualitative interpretation.

Results: Overall Performance

98.17%
Accuracy

97.23%
Precision

97.40%
Recall

97.28%
F1-score

99.58%
Specificity

99.72%
ROC-AUC

Five-fold accuracy ranged from 98.05% to 98.28%, indicating stable performance across folds.

Note: The paper contains differing AUC values in its prose; this slide uses 99.72% from the abstract/conclusion and cross-validation table.

Ablation Study: What Drives Performance?

Accuracy by configuration



Main finding

SupCon pretraining provides the largest observed improvement over the image-only baseline.

Fusion finding

Attention-based fusion adds adaptive weighting; the gain over strong SupCon features is smaller.

Ablation values are reproduced from Table IV; percentages are rounded for readability.

Per-Class Results and Error Patterns

Per-class F1-score (paper-reported)



Best F1

NV: 99.12%

Lowest F1

Melanoma: 95.32%

Main confusion noted: nevi ↔ benign keratosis, which can appear visually similar.

Rare-class metrics should be interpreted alongside sample counts.

Explainability and Discussion

Grad-CAM

Highlights image regions associated with a prediction.

Grad-CAM++

Provides more refined localization of influential regions.

Clinical meaning

Visual maps can support inspection, but are not a complete clinical explanation.

Discussion: key takeaways

A. SupCon is the primary contributor to the large improvement over the baseline.

B. Clinical metadata provides complementary information; attention can adapt its contribution.

C. The model reports strong overall and per-class performance on the benchmark.

D. External validation and prospective evaluation with dermatology specialists remain necessary.

The paper mentions external validation on ISIC 2018; detailed protocol and metrics are not fully presented in the main results tables.

Conclusion & Future Work

FusionClassifier combines discriminative image learning, adaptive multimodal fusion, and visual explanation in a single framework.

Key result

98.17% accuracy

97.28% F1-score

Key contribution

Swin-Small + SupCon + attention fusion

Explainability

Grad-CAM and Grad-CAM++

Future work

i. Validate on broader and more diverse external datasets.

ii. Incorporate richer clinical information and improve fusion strategies.

iii. Conduct prospective evaluation with dermatology specialists.

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Questions & Discussion

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Thank You

Keep Learning, Keep Grinding